

Squeegee a Mirror Panel SOP (1x Panel, in situ)

One episode cleans one wall-mounted mirror panel, in place.

The base is **passive**. It is pushed by hand to the counter under the mirror and locked there. Nothing is carried away to a table. The glass is worked **as found**, dirty from edge to edge. The spray bottle, the cloth, and the squeegee all sit on the counter directly below it.

The order never changes. Spray the glass. Pull the four passes. Detail the four corners. The blade is wiped on the cloth after every pass.

The squeegee comes first and the cloth comes after. Each tool is taken up in its own step, one gripper at a time, and each is taken off its own rest and laid back on it. Through the passes both are off their rests at once, one in each gripper: the squeegee gripper pulls, and the other gripper holds the cloth up in the air for the blade to be wiped on. **The squeegee always goes up first and comes back down first.**

Three rules hold for the whole episode.

Every approach is **face-on**: straight in toward the wall and straight back out, never sideways along the glass.

The mirror, the wall, and the counter are **never loaded**: nothing leans on them, and no push is hard enough to move the panel.

Only the squeegee blade and the cloth touch the glass.

The two arms never cross. The glass is split down the middle by the **panel midline**. The **left gripper works the left half**. The **right gripper works the right half**. Neither gripper crosses the midline while it is touching the glass.

The three tools sit in a short row across the middle of the counter. Each gripper takes a tool from its place and puts it back on its place. **No tool is ever handed from gripper to gripper in the air. Only one gripper is at the tool row at a time.**

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own. It is pushed into place by hand. It is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the counter. Stop it **square to the wall**, so the top edge of the mirror runs straight across the frame and neither side edge of the glass sits nearer than the other.
2. Stop it **centered on the mirror panel**, so the left edge and the right edge of the glass are the same distance out from the middle of the base. The panel midline then lines up with the middle of the base.
3. Stop it **close enough** that both grippers reach the glass face-on without either arm extending. Stop it **far enough** that no arm, wrist, forearm, or part of the base touches the counter, the wall, or the frame when both arms work at once.
4. Check the **left half reach**. With the base parked, the **left gripper** puts the blade flat on the glass at the top-left corner, at the top of the panel midline, and at the bottom edge of the left half. It pulls a full pass from the top edge to the bottom edge in the left half. It does all of this without extending and without the wrist touching the frame or the counter.
5. Check the **right half reach**. With the base parked, the **right gripper** does the same in the right half: blade flat at the top-right corner, at the top of the panel midline, and at the bottom edge of the right half, and a full pass top to bottom.
6. Check the **tool row reach**. With the base parked, each gripper reaches the bottle spot, the cloth rest, and the squeegee rest without extending.
7. Lock or brake the base. Push it firmly once by hand. It must not roll, creep, or turn.
8. If any of lines 1 to 6 fails, push the base to a new park by hand and start again at line 1. If line 4 or line 5 still fails after a re-park, the panel is too wide or too tall to work from one park. Do not work it.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the furniture hard enough to shift it. No hand touches it. It is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the mirror panel. Its frame includes the whole glass from the left edge to the right edge and from the top edge to the bottom edge, the frame all the way round, and the counter with the bottle spot, the cloth rest, and the squeegee rest.
3. The camera reads the glass from the front at an angle where dirt, water beads, runs, and streaks are all visible against the reflection. If the reflection or a glare patch hides part of the glass, move the camera until the whole panel reads.

4. The camera reads the line where the glass meets the frame, on all four sides. You can see whether the blade has run onto the frame.
5. The camera reads the blade from the front. You can see whether water is still on the rubber after a wipe.
6. Both arms are at home with grippers open.
7. The mirror panel is fixed firm to the wall. It does not move, rock, or swing when the blade is pressed on the glass.
8. The counter is fixed. It does not move or rock when a tool is set on it.
9. The **left arm** reaches the whole left half of the glass, from the left edge to the panel midline and from the top edge to the bottom edge, and it reaches the bottle spot, the cloth rest, and the squeegee rest. It reaches all of them without extending to a joint limit.
10. The **right arm** reaches the whole right half of the glass, from the panel midline to the right edge and from the top edge to the bottom edge, and it reaches the same three tool places. It reaches all of them without extending to a joint limit.
11. Each arm comes at the glass **face-on**, straight in toward the wall and straight back out. No part of the arm touches the wall, the frame, or the counter on the way in or out.
12. Neither arm has to reach past the panel midline to do its own half.
13. Both arms meet in the air over the middle of the counter for the blade wipe without touching each other, with one gripper holding the cloth out and the other bringing the blade to it. Neither arm has to pass the other to do it.
14. If a place cannot be reached, re-park the base by the Base positioning steps until lines 9 to 13 hold.

Materials checklist

1. The **mirror panel** is one flat rectangular mirror in a plain flat frame, fixed to the wall above the counter. It is the only mirror in frame.
2. The panel is wide enough that **four passes of the blade with overlap cover it from the left edge to the right edge**, and narrow enough that **two of those passes cover the left half and two cover the right half**. It is short enough that one gripper pulls from the top edge to the bottom edge in one move without extending.
3. The **glass is dirty from edge to edge and from top to bottom**. Visible smears, spots, and finger marks cover the whole panel. The dirt reads on camera from the front.
4. The glass is **dry** at episode start. No water, no beads, no runs anywhere on it.
5. The glass carries no tape, stickers, labels, or paint marks. Nothing is stuck to it and nothing hangs over it.
6. The glass has no chip, crack, or scratch, and no mark in the glass itself that a correct pass will not lift.

7. The **frame** is a plain flat border round the glass. The line where the glass meets the frame reads from the front on all four sides. The frame is dry and clear at episode start.
8. Nothing is mounted on the wall within either arm's reach. Nothing hangs over the top edge of the panel.
9. The **counter** runs directly below the panel. It is level, bare, and dry. It holds three things and nothing else, in a short row across its middle: the **bottle spot** on the left of the row, the **cloth rest** in the middle of the row, and the **squeegee rest** on the right of the row. **Both grippers reach all three without extending.** Nothing on the counter is marked or taped out.
10. The three tool places sit close enough together to be one row and far enough apart that a gripper coming down on one does not touch the tool on either side.
11. The **spray bottle** is a trigger bottle standing upright on its own base on the bottle spot. It holds enough glass cleaner for at least eight bursts.
12. The bottle's **body and trigger both sit inside one gripper's opening**, so **one gripper works the bottle on its own**. Closing the gripper on the body squeezes the trigger and fires a burst. Opening it lets the trigger spring back. The second arm is never needed to fire it.
13. The trigger springs back on its own as soon as the gripper opens, so no liquid carries on after a burst.
14. The bottle does not tip when a gripper closes on it. One gripper carries it on its own without help.
15. The bottle's nozzle is **already set to a fan spray** and is never turned during the episode. One burst lands a wet patch on the glass, not a drifting mist.
16. The **squeegee** is a single straight blade on a rigid handle. Its rubber is clean, straight, and free of nicks, waves, and tears.
17. The blade is wide enough that two passes with overlap cover one half of the panel. It is no wider than half the panel.
18. The squeegee lies flat on the squeegee rest with its **handle toward the front of the counter** and its **blade level, rubber facing up**. Either gripper closes on the handle straight in from the front without touching the rubber.
19. The handle sits inside one gripper's opening. The squeegee does not turn or slip in the grip when the blade is pressed on the glass.
20. The **cloth** is clean, dry, and folded flat. It lies flat on the cloth rest in the middle of the tool row. The rest holds nothing else. Held up by one edge in a gripper, it keeps its fold: it does not flop open, hang loose, or slip out under the pressure of a blade wipe.
21. The cloth is absorbent enough to take the water off the blade in two strokes without dripping. It does not shed fibres onto the glass.
22. The counter has clear space above all three tool places, so each gripper comes down onto a place straight from the front.

23. The floor under the base is level, dry, and clear.

Workspace layout

Nothing anywhere is marked or taped out. You judge every place below by eye against the mirror and the counter themselves: the edges of the glass, the frame, and the tools already in place.

- **Wall:** the surface the mirror hangs on. It is never touched, never leaned on, and never pushed.
- **Mirror panel:** the unit being cleaned, the glass and the frame round it. It is never moved.
- **Glass:** the mirror surface itself. Only the squeegee blade and the cloth ever touch it.
- **Frame:** the flat border round the glass. Nothing is dragged onto it and nothing rests on it.
- **Top edge / bottom edge / left edge / right edge:** the four lines where the glass meets the frame.
- **Corners:** the four places where two edges of the glass meet: top-left, top-right, bottom-left, bottom-right.
- **Panel midline:** the line halfway across the glass, judged by eye. It splits the panel into a **left half** and a **right half**. It is the boundary between the two arms.
- **Left half:** the glass from the left edge to the panel midline. **Left gripper only.**
- **Right half:** the glass from the panel midline to the right edge. **Right gripper only.**
- **Counter:** the surface directly below the panel. It carries the tool row across its middle. It is never leaned on.
- **Tool row:** the three tool places in a short row across the middle of the counter. Left to right they are the bottle spot, the cloth rest, and the squeegee rest. Both grippers reach all three.
- **Bottle spot:** the place in the tool row where the spray bottle stands at episode start and at episode end. It is the only place the bottle is set down. Both grippers use it, one at a time.
- **Cloth rest:** the place in the middle of the tool row where the folded cloth lies. The cloth is taken up from it and laid back on it three times in an episode: once by the right gripper for the left-half passes, once by the left gripper for the right-half passes, and once by each gripper in turn for the corners. It is the only place either gripper sets the cloth down, and only one gripper ever holds it.
- **Squeegee rest:** the place in the tool row where the squeegee lies at episode start and at episode end. It is the only place the squeegee is set down. Both grippers use it, one at a time.

Arm lanes

The two arms never cross. This holds for the whole episode.

- The **left gripper works the left half of the glass**. The **right gripper works the right half**. Neither gripper crosses the panel midline while it is touching the glass.

- The **left gripper always stays left of the right gripper**. Neither arm reaches over, under, around, or past the other. Neither arm reaches across the front of the other arm's body.
- **Only one arm is at the tool row at a time**. There is no exception. While one gripper is taking a tool off the row or laying one back, the other stays clear of the row, either out at the front or holding its own tool in the air.
- **The two arms meet only in the air, for the blade wipe**. One gripper holds the cloth in the **wipe position** and the other brings the blade to it. Even then the left gripper stays left of the right gripper and neither arm passes the other.
- The bottle, the cloth, and the squeegee pass between the grippers only by being set down on their own place and taken up again. **Nothing is ever handed from gripper to gripper in the air**.

Arm assignments

- **Left gripper**. Fires the four bursts on the left half. Pulls pass 1 and pass 2 and wipes the blade. Holds the cloth in the wipe position while the right gripper pulls pass 3 and pass 4. Details the top-left and bottom-left corners. It never touches the right half of the glass.
- **Right gripper**. Fires the four bursts on the right half. Holds the cloth in the wipe position while the left gripper pulls pass 1 and pass 2. Pulls pass 3 and pass 4 and wipes the blade. Details the top-right and bottom-right corners. It never touches the left half of the glass.
- **One gripper never holds both tools**. The squeegee is in one gripper and the cloth in the other, or one of them is on its rest.
- Only one arm moves a tool to or from the tool row at a time.

Vocabulary

- **Face-on approach**: the gripper comes at the glass straight in toward the wall and leaves straight back out. It never travels sideways along the wall while touching the glass. It never swings across the panel. Every action on the glass is a face-on approach.
- **Blade**: the rubber strip along the bottom of the squeegee. **Rubber** is the working edge of it, the thin lip that meets the glass. **Handle** is the rigid grip above the blade. A gripper closes on the handle only.
- **Blade flat**: the whole length of the rubber touches the glass at once. No end is lifted. No gap shows under any part of it.
- **Blade level**: the blade runs straight across, side to side, so both its ends sit at the same height on the glass.
- **Pass**: one pull of the blade from the top edge of the glass straight down to the bottom edge. The blade stays flat and level the whole way, in one steady move.
- **Steady pull**: the blade moves down at one speed. It does not stop, jerk, speed up, or change angle.

- **Dry strip:** the band of glass a finished pass has left behind. It reads clear from the front, against the wet glass beside it.
- **Wet band:** a stretch of glass between two dry strips, or at an edge, that still shows water. It is what the pass check looks for.
- **Overlap:** the new pass is set so that **about a third of the blade sits on the dry strip** the last pass left. The rest sits on wet glass. You judge it by eye against the line between wet and dry.
- **Wipe position:** where the cloth is held in the air for a blade wipe. The gripper holding it grips it at its **outer edge**, the edge on that gripper's own side, and holds it out over the middle of the counter: folded, flat, level, still, well clear of the counter surface, and clear of the glass, the frame, and the wall. The free face of the cloth lies on the other gripper's side of the grip, so the blade is brought to it without either arm passing the other.
- **Wipe the blade:** the gripper holding the squeegee brings the rubber to the cloth held in the **wipe position** and drags it across the free face of the cloth from one end of the blade to the other, once each way, so the rubber comes off the cloth with no water on it. The gripper holding the cloth holds it still and takes the pressure.
- **Clear:** no beads, runs, standing water, or smears are visible on that stretch of glass when it is read from the front. This is what a finished pass and a finished corner both look like.
- **Detail a corner:** press the folded cloth into the corner where the two edges of the glass meet. Draw it along each of those two edges a short way back onto the glass, so the corner and the glass just beside it read clear. It is a wipe of the corner, not a wipe of the whole panel.
- **Spray place:** one of the eight named places on the glass a burst is aimed at in Step 1. They are not marked. You read them by eye against the edges of the glass and the panel midline.
- **Burst:** the liquid one squeeze of the trigger throws at the glass. It leaves a **wet patch**, the wet mark a burst lands on the glass.
- **Loaded:** pressed, leaned on, or pushed hard enough to move. The mirror, the wall, and the counter are never loaded.

Steps

Step 1: Spray the glass

Goal: the top two-thirds of the glass carries eight wet patches, four on each half. The bottle is back upright on its spot.

Each gripper sprays its own half. The bottle is worked by **one gripper on its own**. Closing the gripper on the body squeezes the trigger. The bottle is held in open air in front of the panel. **The nozzle never touches the glass.** No burst lands on the frame, the wall, or the counter.

1.1 Spray the left half

- The **left gripper** comes in from the front and closes on the spray bottle **around its body**, just hard enough to hold it without firing.
- Lift the bottle straight up off the bottle spot. Keep it **upright** the whole time. Do not tip it. Do not turn it. Do not carry it over the cloth rest or the squeegee rest.
- Carry it out in front of the **left half of the glass**. Hold it far enough back that one burst lands a wet patch and not a drifting mist, with the nozzle pointing level at the glass.
- Fire **four bursts**, one wet patch per burst, in this order:
 - **top third of the left half**: near the left edge, then just left of the panel midline;
 - **middle third of the left half**: near the left edge, then just left of the panel midline.
- Squeeze to fire, then open to let the trigger spring back, then move to the next spray place. Do not move the bottle while the trigger is squeezed.
- Keep the nozzle pointed inside the left half for every burst. Never aim toward the frame, the wall, the counter, the cloth, the squeegee, a camera, or the right half. Never fire while the nozzle is turned away from the panel.
- Take the bottle straight back to the bottle spot. Stand it upright on its own base. Open the **left gripper** and draw the arm out to the front.

Check: the **left gripper** is clear of the bottle. The bottle stands upright on its spot. Four wet patches are visible on the left half. No liquid has landed on the frame, the wall, or the counter. If a spray place was missed, take the bottle up again with the **left gripper** and fire one burst at that place.

1.2 Spray the right half

- The **right gripper** takes the bottle off the bottle spot the same way, closing on its body around it.
- Carry it out in front of the **right half of the glass**, upright, nozzle level.
- Fire **four bursts**, one wet patch per burst, in this order:
 - **top third of the right half**: just right of the panel midline, then near the right edge;
 - **middle third of the right half**: just right of the panel midline, then near the right edge.
- Keep the nozzle pointed inside the right half for every burst, under the same rules as Step 1.1.
- Take the bottle straight back to the bottle spot. Stand it upright on its own base. Open the **right gripper** and draw the arm out to the front.

Check: both grippers are clear of the bottle. The bottle stands upright on its spot. Eight wet patches are visible on the glass, four on each half. No liquid has landed on the frame, the wall, or the counter. If a spray place was missed, take the bottle up again with the **right gripper** and fire one burst at that place.

Expected state: dirty glass carrying eight wet patches over its top two-thirds, water starting to run down. Bottle upright on its spot, squeegee on its rest, cloth flat on its rest, both arms clear.

Step 2: Pull the two left-half passes

Goal: the left half of the glass is pulled clear in two top-to-bottom passes. The blade is wiped after each one.

The **left gripper** holds the squeegee for the whole of this step. The **right gripper** holds the cloth for the whole of it: it takes the cloth off its rest, holds it still in the **wipe position**, and lays it back at the end. **Both tools are off their rests at once, one in each gripper.** They are taken up one at a time and in order: the squeegee first, then the cloth. They go back the same way, squeegee first.

2.1 Take up the squeegee

- With the **left gripper**, come in from the front and close on the squeegee **handle**. Do not touch the rubber.
- Lift it straight up off the squeegee rest. Hold it with the **blade level and the rubber facing the glass**.
- Keep the rubber clear of the counter, the frame, and the wall. Nothing but the glass and the cloth ever touches it.
- Draw the arm out to the front, clear of the tool row, so the other arm can come in.

Check: the **left gripper** holds the handle. The rubber is untouched and the blade is level. The **right gripper** is still clear at the front and the tool row is free.

2.2 Take up the cloth

- Only once the squeegee is up and clear of the tool row: with the **right gripper**, come in from the front and close on the folded cloth on the cloth rest, gripping it at its **right edge**.
- Lift it straight up off the rest, folded and flat.
- Bring it to the **wipe position** and hold it still there.
- Hold it there for the whole of the step. Do not chase the blade with it. Do not carry it to the glass. Do not let it unfold, hang open, or trail on the counter.

Check: the **right gripper** holds the cloth folded, flat and still in the wipe position, clear of the counter, the frame, the wall, and the glass. The **left gripper** is left of the **right gripper**.

2.3 Set the blade at the top of the pass

- With the **left gripper**, bring the blade at the glass **face-on**. Set it flat against the glass at the **top edge**, blade level, rubber touching the glass along its whole length.
- Set it in the right place for the pass:
 - **Pass 1:** the **left end of the blade is level with the left edge of the glass**.

- **Pass 2:** about a **third of the blade sits on the dry strip** pass 1 left. The rest sits on wet glass. The **right end of the blade reaches the panel midline** and goes no further.
- Keep the rubber just inside the top edge. Do not run it up onto the frame. Do not let either end run past a side edge onto the frame, and do not let the right end cross the panel midline.

Check: the blade sits flat and level against the glass, in the place named for this pass, with the rubber fully on the glass and clear of the frame. If an end is lifted, or the blade is on the frame, or the right end is past the midline, take the blade straight off the glass toward the front and set it again.

2.4 Pull the pass

- With the **left gripper**, pull the blade **straight down** from the top edge to the bottom edge in one **steady pull**. Keep it flat and level and pressed on the glass the whole way.
- Press only hard enough that the rubber stays flat on the glass. Do not press hard enough to bend the frame, move the panel, or push the mirror against the wall.
- Do not lift the blade partway. Do not stop mid-pass. Do not tilt it. Do not slide it sideways.
- Stop when the rubber reaches the **bottom edge** of the glass. Do not run it down onto the frame.
- Take the blade **straight off the glass toward the front**, away from the wall. Never swing it sideways across the glass to get back to the top.

Check: the strip the blade has just pulled reads **clear** from the front, top to bottom. No wet band is left in it and no line runs down it. If a wet band or a line shows, wipe the blade by Step 2.5 and pull that same pass again, once.

2.5 Wipe the blade

- The **right gripper** holds the cloth still in the **wipe position**. It does not move to meet the blade, it does not chase it, and it does not wipe the blade itself.
- With the **left gripper**, carry the squeegee to the cloth, keeping the rubber clear of the counter, the frame, and the wall.
- Press the rubber onto the free face of the cloth and drag it across the cloth from one end of the blade to the other, **once each way**.
- Press only hard enough to take the water off. The **right gripper** takes that pressure and holds the cloth still. Do not press hard enough to pull the cloth out of the grip, fold it over, or push the right arm back.
- Wipe on the cloth only. Never wipe the blade on the counter, the frame, the glass, a gripper, or an arm.
- Lift the blade straight off the cloth. The **left gripper** stays left of the **right gripper** the whole time.

Check: the **left gripper** holds the blade clear of the cloth and no water is visible on the rubber. The cloth is still folded flat and still held in the wipe position. If water is still on the rubber, wipe it once each way again. If the cloth has unfolded, slipped in the grip, or come out of it, lay the squeegee back on the squeegee rest and the cloth flat on the cloth rest, then take them both up again by Steps 2.1 and 2.2.

2.6 Put the squeegee back

- Run Steps 2.3 to 2.5 for **pass 1** and then for **pass 2**.
- After the pass 2 wipe, carry the squeegee down to the squeegee rest with the **left gripper** and lay it flat, **handle toward the front** and **blade level, rubber facing up**.
- Open the **left gripper** and draw the arm out to the front.
- The **right gripper** goes on holding the cloth still in the wipe position while this happens. It does not come to the tool row.

Check: the left half of the glass reads clear from the left edge to the panel midline and from the top edge to the bottom edge, apart from the two left corners and the edges. No wet band is left between pass 1 and pass 2, and none at the left edge. The squeegee lies on its rest, handle to the front and rubber up.

2.7 Put the cloth back

- Only once the squeegee is on its rest and the **left gripper** is drawn out clear of the tool row: with the **right gripper**, carry the cloth down to the **cloth rest** and lay it flat and folded on it, the way it started.
- Open the **right gripper** and draw the arm out to the front.

Check: the cloth lies flat and folded on its rest. Both grippers are clear at the front and hold nothing.

Expected state: left half clear, right half still wet, squeegee on its rest, cloth flat on its rest, bottle upright on its spot.

Step 3: Pull the two right-half passes

Goal: the right half of the glass is pulled clear in two top-to-bottom passes. The blade is wiped after each one.

The two grippers swap jobs for this step. The **right gripper** holds the squeegee for the whole of it. The **left gripper** holds the cloth for the whole of it. Both tools are taken up off their rests again, one at a time and in the same order: the squeegee first, then the cloth. **Neither tool is ever handed across in the air.** Each was laid on its own rest at the end of Step 2 and is taken up from there.

3.1 Take up the squeegee

- With the **right gripper**, come in from the front and close on the squeegee **handle**. Do not touch the rubber.
- Lift it straight up off the squeegee rest. Hold it with the **blade level and the rubber facing the glass**.
- Keep the rubber clear of the counter, the frame, and the wall.
- Draw the arm out to the front, clear of the tool row.

Check: the **right gripper** holds the handle, the rubber is untouched and the blade is level, and the tool row is free.

3.2 Take up the cloth

- Only once the squeegee is up and clear of the tool row: with the **left gripper**, come in from the front and close on the folded cloth on the cloth rest, gripping it at its **left edge**.
- Lift it straight up off the rest, folded and flat, and bring it to the **wipe position**.
- Hold it still there for the whole of the step, under the same rules as Step 2.2.

Check: the **left gripper** holds the cloth folded, flat and still in the wipe position, clear of the counter, the frame, the wall, and the glass. The **left gripper** is left of the **right gripper**.

3.3 Set the blade at the top of the pass

- With the **right gripper**, bring the blade at the glass **face-on**. Set it flat against the glass at the **top edge**, blade level, rubber touching the glass along its whole length.
- Set it in the right place for the pass:
 - **Pass 3:** about a **third of the blade sits on the dry strip** pass 2 left, and the rest sits on wet glass. Its **left end is on the dry strip just left of the panel midline**.
 - **Pass 4:** the **right end of the blade is level with the right edge of the glass**, and about a third of the blade sits on the dry strip pass 3 left.
- Keep the rubber just inside the top edge. Do not run it up onto the frame. Do not let either end run past a side edge onto the frame.

Check: the blade sits flat and level against the glass, in the place named for this pass, with the rubber fully on the glass and clear of the frame. If an end is lifted or the blade is on the frame, take the blade straight off the glass toward the front and set it again.

3.4 Pull the pass

- Pull the pass with the **right gripper** under exactly the same rules as Step 2.4: straight down from the top edge to the bottom edge, one steady pull, blade flat and level, then straight off the glass toward the front.

Check: the strip the blade has just pulled reads **clear** from the front, top to bottom. If a wet band or a line shows, wipe the blade by Step 3.5 and pull that same pass again, once.

3.5 Wipe the blade

- The **left gripper** holds the cloth still in the **wipe position**. It does not move to meet the blade and it does not wipe the blade itself.
- With the **right gripper**, carry the squeegee to the cloth and drag the rubber across the free face of the cloth from one end of the blade to the other, **once each way**, under the same rules as Step 2.5.
- Lift the blade straight off the cloth. The **left gripper** stays left of the **right gripper** the whole time.

Check: the **right gripper** holds the blade clear of the cloth and no water is visible on the rubber. The cloth is still folded flat and still held in the wipe position. If the cloth has unfolded, slipped in the grip, or come out of it, lay the squeegee back on the squeegee rest and the cloth flat on the cloth rest, then take them both up again by Steps 3.1 and 3.2.

3.6 Put the squeegee back

- Run Steps 3.3 to 3.5 for **pass 3** and then for **pass 4**.
- After the pass 4 wipe, carry the squeegee down to the squeegee rest with the **right gripper** and lay it flat, **handle toward the front** and **blade level, rubber facing up**.
- Open the **right gripper** and draw the arm out to the front.
- The **left gripper** goes on holding the cloth still in the wipe position while this happens.

Check: read the whole panel from the front. The glass reads clear from the left edge to the right edge and from the top edge to the bottom edge, apart from the four corners and the edges. No wet band shows between any two passes and none at either side edge.

If a wet band shows, pull one more pass over it before the cloth goes back. If it is in the **left half**, lay the cloth back on its rest with the **left gripper**, then take the squeegee with the **left gripper** and the cloth with the **right gripper**, run that one pass by Steps 2.3 to 2.5, and lay both tools back, squeegee first. If it is in the **right half**, take the squeegee up again with the **right gripper**, run that one pass by Steps 3.3 to 3.5 with the cloth still held in the **left gripper**, and lay the squeegee back.

3.7 Put the cloth back

- Only once the squeegee is on its rest and the **right gripper** is drawn out clear of the tool row: with the **left gripper**, carry the cloth down to the **cloth rest** and lay it flat and folded on it.
- Open the **left gripper** and draw the arm out to the front.

Check: the cloth lies flat and folded on its rest. Both grippers are clear at the front and hold nothing.

Expected state: whole glass clear apart from the corners and edges. Squeegee on its rest, handle to the front and rubber up. Cloth flat on its rest. Bottle upright on its spot.

Step 4: Detail the four corners

Goal: all four corners of the glass read clear. The cloth is back on its rest.

Both tools are back on their rests before this step starts. **The squeegee is not touched again.** From here the cloth is the only tool.

Each gripper details the two corners on its own side. It takes the cloth up at the start of its turn and lays it flat back on the rest at the end of it, before the other gripper takes it. **The two grippers never hold the cloth at the same time.**

The top corner is always done before the bottom corner on that side, so water drawn off the top lands on glass that has not been detailed yet.

4.1 Take up the cloth with the left gripper

- Check the squeegee is lying on its rest and the **right gripper** is clear at the front.
- With the **left gripper**, come in from the front and close on the folded cloth on the cloth rest.
- Lift it straight up off the rest. Keep it folded and flat. Do not let it unfold, hang open, or trail on the counter.

Check: the **left gripper** holds the cloth folded and flat, clear of the counter. The **right gripper** is clear at the front and holds nothing.

4.2 Detail the two left corners

- With the **left gripper**, bring the cloth at the glass **face-on** and press it into the **top-left corner**.
- Draw it a short way along the **top edge** toward the middle, then back into the corner. Then draw it a short way down the **left edge**, then back into the corner. Keep the cloth pressed flat on the glass the whole time.
- Take the cloth **straight off the glass toward the front**.
- Do the same at the **bottom-left corner**. Press into the corner. Draw a short way along the **bottom edge** toward the middle and back. Then draw a short way up the **left edge** and back.
- Do not drag the cloth onto the frame. Do not wipe the frame. Do not lean the wrist or the forearm on the glass or the counter. Do not carry the cloth past the panel midline.

Check: the top-left corner, the bottom-left corner, and the glass just beside them read **clear** from the front. If a corner is still wet, streaked, or missed, detail that corner again, once, with the

cloth still in the **left gripper**.

4.3 Put the cloth back

- With the **left gripper**, carry the cloth down to the **cloth rest** and lay it flat and folded on it, the way it started.
- Open the **left gripper** and draw the arm out to the front.

Check: the cloth lies flat and folded on its rest. The **left gripper** is clear at the front and holds nothing.

4.4 Take up the cloth with the right gripper

- Check the **left gripper** is clear at the front.
- With the **right gripper**, come in from the front and close on the folded cloth on the cloth rest.
- Lift it straight up off the rest, folded and flat.

Check: the **right gripper** holds the cloth folded and flat, clear of the counter. The **left gripper** is clear at the front and holds nothing.

4.5 Detail the two right corners

- With the **right gripper**, bring the cloth at the glass **face-on** and press it into the **top-right corner**.
- Draw it a short way along the **top edge** toward the middle, then back into the corner. Then draw it a short way down the **right edge**, then back into the corner.
- Take the cloth **straight off the glass toward the front**.
- Do the same at the **bottom-right corner**. Press into the corner. Draw a short way along the **bottom edge** toward the middle and back. Then draw a short way up the **right edge** and back.
- Do not drag the cloth onto the frame. Do not carry the cloth past the panel midline.

Check: the top-right corner, the bottom-right corner, and the glass just beside them read **clear** from the front. If a corner is still wet, streaked, or missed, detail that corner again, once, with the cloth still in the **right gripper**.

4.6 Put the cloth back

- With the **right gripper**, carry the cloth down to the **cloth rest** and lay it flat and folded on it.
- Open the **right gripper** and draw the arm out to the front.

Check: the cloth lies flat and folded on its rest. Both grippers are clear at the front and hold nothing.

Expected state: the whole glass reads clear from edge to edge and corner to corner. The frame is dry and unmarked. The bottle stands upright on its spot. The squeegee lies on its rest,

handle to the front and rubber up. The cloth lies flat on its rest.

Step 5: End the episode

- Confirm the glass reads clear edge to edge and corner to corner. No wet band, run, bead, or line is left on it.
- Confirm the frame is dry, the panel hangs as it did, and the base has not moved.
- Confirm the counter is dry and holds the bottle upright on its spot, the squeegee on its rest, and the cloth on its rest, and nothing else.
- Correct any failed check before ending.
- Return both arms home with grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Stand the spray bottle upright on the bottle spot. Top up the cleaner to enough for at least eight bursts. Check the nozzle is still set to a fan spray and the trigger springs back on its own.
2. Lay the squeegee flat on the squeegee rest with its handle toward the front and its blade level, rubber facing up.
3. Put a clean, dry, flat-folded cloth on the cloth rest and take the used one away.
4. Dry the glass fully. Then put fresh smears, spots, and finger marks on it from edge to edge and from top to bottom. Check it reads on camera from the front.
5. Dry the frame, the counter, and the wall below the panel. Wipe up any water on the floor.
6. Replace a squeegee whose rubber is nicked, wavy, torn, or that leaves a line on the glass under a correct pass.
7. Replace a cloth that is wet, shedding, or that will no longer take the water off the blade in two strokes.
8. Replace a bottle that will not spray a fan, whose body and trigger no longer sit inside one gripper's opening, whose trigger no longer springs back on its own, or that tips when a gripper closes on it.
9. Check the mirror panel is still fixed firm to the wall, hangs square, and does not rock when the glass is pressed.
10. Check the counter is fixed, level, and clear of everything but the three tool places.
11. Check the base is still locked and parked square. Then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the mirror shifts in frame, the panel changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 5. The base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Arms crossed

- **Visible cue:** the **left gripper** touches the glass right of the panel midline, or carries a tool past the midline. The **right gripper** touches the glass left of the panel midline, or carries a tool past it. The left gripper goes to the right of the right gripper. One arm reaches over, under, around, or past the other, or across the front of the other arm's body, including on the way to a blade wipe. Or both arms are at the tool row at once.
- **SOP rule broken:** Arm lanes and Arm assignments. The left gripper works the left half, the right gripper works the right half, and the left gripper always stays left of the right gripper.
- **Coaching note:** the panel midline is a wall. If you have to reach past it, or past the other arm, the wrong arm is doing the job.

Violation: Not a face-on approach

- **Visible cue:** a gripper comes at the glass from the side or along the wall instead of straight in from the front. It travels sideways along the glass outside the prescribed pull. Or it swings the blade or the cloth across the panel instead of taking it straight off toward the front.
- **SOP rule broken:** Steps 1 to 4. Every action on the glass is a face-on approach: straight in toward the wall and straight back out.
- **Coaching note:** straight in, do the work, straight out. Nothing travels sideways while it is touching the glass.

Violation: Mirror, wall, or counter loaded

- **Visible cue:** the panel moves, rocks, or shifts on the wall. A gripper, wrist, or forearm leans or rests on the glass, the frame, the wall, or the counter. Or the blade is pressed hard enough to push the mirror back.
- **SOP rule broken:** Steps 1 to 4. The mirror, the wall, and the counter are never loaded, and the blade is pressed only hard enough to keep the rubber flat.
- **Coaching note:** the panel holds nothing up for you. Light pressure, and let the arm carry its own weight.

Violation: Something other than the blade or the cloth touched the glass

- **Visible cue:** a gripper finger, a wrist, a forearm, the squeegee handle, or the bottle nozzle touches the glass at any point.
- **SOP rule broken:** Steps 1 to 4. Only the squeegee blade and the cloth ever touch the glass.
- **Coaching note:** two things reach the glass and nothing else. Everything else stays back in open air.

Violation: Wrong order

- **Visible cue:** the squeegee is taken up before both halves are sprayed. A pass is pulled on dry glass. A pass is pulled before the cloth is up in the wipe position. The right-half passes are pulled before the left-half passes. A corner is detailed before all four passes are done. Or a step is skipped.
- **SOP rule broken:** Steps 1 to 4. Spray both halves, pull the two left passes, pull the two right passes, then detail the corners.
- **Coaching note:** spray, left passes, right passes, corners. Nothing is pulled dry and nothing is detailed early.

Violation: Spray aimed wrong

- **Visible cue:** a burst is aimed at or lands on the frame, the wall, the counter, the cloth, the squeegee, a camera, or the floor. The nozzle points away from the glass while the trigger is squeezed. The nozzle is held against the glass. The gripper tips the bottle past level or turns the nozzle. Or the gripper moves the bottle while the trigger is still squeezed.
- **SOP rule broken:** Step 1. Keep the bottle upright with the nozzle pointed level inside your own half of the glass for every burst. Open the gripper before moving the bottle.
- **Coaching note:** nozzle level and inside your half before you squeeze. Open before you move. Liquid lands on the glass, never on a cloth or the frame.

Violation: Wrong number of bursts, or bursts in the wrong places

- **Visible cue:** fewer or more than four bursts are fired on a half. Or the wet patches do not land two across the top third and two across the middle third of that half, one near the outer edge and one near the panel midline in each row.

- **SOP rule broken:** Step 1. Fire four bursts on each half, two across the top third and two across the middle third, in that order.
- **Coaching note:** four bursts a side, two rows of two, outer edge then midline. Read the patches before the bottle goes back.

Violation: Bottle handled or put back wrong

- **Visible cue:** the gripper closes on the bottle above the body instead of around it. It tips or turns the bottle. It carries it over the cloth rest or the squeegee rest. It sets it down somewhere other than the bottle spot. Or it leaves it lying on its side or standing off its base.
- **SOP rule broken:** Step 1. Close on the body, lift the bottle straight up, keep it upright, carry it clear of the other tool places, and stand it back upright on its own base on the bottle spot.
- **Coaching note:** body in the grip, upright out, upright back.

Violation: Squeegee taken up, held, or put back wrong

- **Visible cue:** the gripper closes on the rubber or the blade instead of the handle. It holds the blade tilted or not level. It lets the squeegee turn or slip in the grip. It rests the rubber on the counter, the frame, or the wall. Or it lays the squeegee back rubber down, handle to the back, or anywhere other than the squeegee rest.
- **SOP rule broken:** Steps 2.1, 2.6, 3.1, and 3.6. Close on the handle only, hold the blade level with the rubber facing the glass, keep the rubber clear of everything but the glass and the cloth, and lay it back handle to the front and rubber up.
- **Coaching note:** handle only, blade level, rubber up when it goes down. The rubber is what does the work, and nothing else touches it.

Violation: Squeegee handed across in the air

- **Visible cue:** the squeegee passes from one gripper to the other in mid-air, or one gripper takes it while the other is still holding it, instead of it being laid on the squeegee rest and taken up again.
- **SOP rule broken:** Arm lanes, Step 2.6, and Step 3.1. The squeegee passes between grippers only by being laid on the squeegee rest.
- **Coaching note:** down on the rest, hand clear, then the other hand takes it. Nothing crosses in the air.

Violation: Pass not set right at the top

- **Visible cue:** the blade starts below the top edge. It starts tilted or with one end lifted off the glass. It starts with an end on the frame. Or it starts in the wrong place for that pass: pass 1 not at the left edge, pass 2 past the panel midline, or pass 4 not reaching the right edge.
- **SOP rule broken:** Steps 2.3 and 3.3. Set the blade flat and level at the top edge, in the place named for that pass.
- **Coaching note:** set it, look at it, then pull. A pass that starts wrong cannot finish right.

Violation: Pass run wrong

- **Visible cue:** the blade lifts off partway down. It stops mid-pass. It jerks or changes speed. It tilts, or is slid sideways during the pull. Or the blade is swung sideways across the glass to get back to the top instead of coming straight off toward the front.
- **SOP rule broken:** Steps 2.4 and 3.4. Pull straight down from the top edge to the bottom edge in one steady pull, then take the blade straight off the glass toward the front.
- **Coaching note:** one steady pull, all the way down, then straight out. Stopping mid-pass leaves a line you cannot pull out.

Violation: Blade run onto the frame or off an edge

- **Visible cue:** the rubber runs up onto the frame above the top edge, or down onto the frame below the bottom edge. An end of the blade runs past a side edge onto the frame. Or water is drawn onto the frame and left there.
- **SOP rule broken:** Steps 2.3, 2.4, 3.3, and 3.4. Keep the rubber just inside the top edge, stop it at the bottom edge, and keep both ends of the blade on the glass.
- **Coaching note:** the glass is the whole run. The frame is not part of the pass.

Violation: Passes not overlapped

- **Visible cue:** a wet band is left between two dry strips. A pass is set clear of the strip before it. Or a pass is set so far onto the dry strip that it covers almost no new glass.
- **SOP rule broken:** Steps 2.3 and 3.3. Set each new pass so about a third of the blade sits on the dry strip the last pass left.
- **Coaching note:** a third on the dry, the rest on the wet. Read the wet line before you set the blade.

Violation: Passes worked in the wrong direction or a strip left out

- **Visible cue:** the passes run right to left. They run out of order. Fewer than four passes are pulled. Or the glass at a side edge is never pulled.
- **SOP rule broken:** Steps 2 and 3. Run four passes from the left edge rightward: pass 1 at the left edge, pass 2 to the panel midline, pass 3 from the midline, pass 4 reaching the right edge.
- **Coaching note:** left edge first, work right, finish on the right edge. That way the water is always pushed onto glass still to do.

Violation: Blade not wiped after a pass, or wiped on the wrong thing

- **Visible cue:** a gripper goes straight from the bottom of one pass to the top of the next without going to the cloth. The wipe is one stroke instead of one each way. Water is still visible on the rubber when the next pass starts. Or the blade is wiped on the counter, the frame, the glass, a gripper, or an arm.

- **SOP rule broken:** Steps 2.5 and 3.5. Carry the blade to the held cloth after every pass and drag the rubber across it once each way, on the cloth only.
- **Coaching note:** every pass ends at the cloth. A wet blade puts the dirt straight back on the glass.

Violation: Tools taken up or put back out of order

- **Visible cue:** the cloth is lifted off its rest before the squeegee is up and clear of the tool row. The cloth is laid back before the squeegee is on its rest. Both grippers are at the tool row at once. Or one gripper holds the squeegee and the cloth at the same time.
- **SOP rule broken:** Steps 2.1, 2.2, 2.6, 2.7, 3.1, 3.2, 3.6, and 3.7, and Arm lanes. The squeegee goes up first and comes back down first. Each tool is taken up in its own step, by one gripper, off its own rest, with the other arm clear of the row.
- **Coaching note:** squeegee up, then cloth. Squeegee down, then cloth. One hand at the row at a time.

Violation: Cloth held wrong for the wipe

- **Visible cue:** the cloth is held by its inner edge, so the arm holding it has to reach past the other arm. It is held low against the counter, or against the frame, the wall, or the glass. It is held loose, unfolded, or hanging open. The holding gripper moves to meet the blade, chases it, or wipes the blade itself. Or it lets the cloth slip in the grip, fold over, or come out of the grip during the wipe.
- **SOP rule broken:** Steps 2.2, 2.5, 3.2, and 3.5. Grip the cloth at its outer edge and hold it out over the middle of the counter, folded, flat, level, still and clear of everything. The blade comes to the cloth.
- **Coaching note:** the cloth is a target, not a tool. Hold it out on your own edge, hold it still, and let the blade come to it.

Violation: Corners detailed out of order

- **Visible cue:** the bottom corner on a side is detailed before the top corner on that side. Or a corner is detailed before all four passes are done and the squeegee is back on its rest.
- **SOP rule broken:** Step 4. Finish the passes, put the squeegee back, then top corner before bottom corner on each side.
- **Coaching note:** top before bottom, always. Water only ever runs one way.

Violation: Corner left wet, streaked, or missed

- **Visible cue:** a corner still shows water, a bead, or a streak after the cloth has left it. The cloth is drawn past the corner without being pressed into it. Or a corner is never touched at all.
- **SOP rule broken:** Steps 4.2 and 4.5. Press the cloth into each corner and draw it a short way along both edges and back, until the corner reads clear.

- **Coaching note:** into the corner first, then out along each edge. A swipe past the corner leaves the corner.

Violation: Cloth handed across or set down wrong

- **Visible cue:** the cloth passes from one gripper to the other in mid-air, or one gripper takes it while the other is still holding it. It is set down anywhere other than the cloth rest. It is left bunched, folded over, or hanging off the rest. Or it is dragged onto the frame or laid on the glass.
- **SOP rule broken:** Arm lanes and Steps 2.2, 2.7, 3.2, 3.7, 4.1, 4.3, 4.4, and 4.6. The cloth passes between grippers only by being laid flat on the cloth rest and taken up again.
- **Coaching note:** down on the rest and up again, never across, and always flat.

Violation: Tools not returned to their places

- **Visible cue:** at the end of the episode the bottle is off its spot or not upright, the squeegee is off its rest or lying with its rubber down or its handle to the back, or the cloth is off its rest.
- **SOP rule broken:** Steps 1, 2.6, 2.7, 3.6, 3.7, and 4. The bottle stands upright on the bottle spot, the squeegee lies on its rest handle to the front and rubber up, and the cloth lies flat on the cloth rest.
- **Coaching note:** three tools, three places, exactly as they started.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped. Or a check is made and the fault it finds is left uncorrected: a missed spray place, a blade set on the frame, a wet band left between passes, water left on the rubber, or a corner left wet.
- **SOP rule broken:** Steps 1, 2, 3, and 4. Run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped or knocked over

- **Visible cue:** the bottle, the squeegee, or the cloth is dropped, knocked over, or ends up on the floor. The bottle is knocked off its spot. Or a tool is swept off the counter by an arm on its way in or out.
- **SOP rule broken:** Steps 1 to 4. Nothing is dropped or knocked out of its place, and every arm comes in and goes out clear of the counter.
- **Coaching note:** check the path and the landing place before the arm moves, and set tools down before letting go.

Violation: Wrong arm used

- **Visible cue:** the **left gripper** sprays, pulls a pass on, or details a corner of the right half. The **right gripper** does any of those on the left half. A gripper holds two tools at once. Or an

object is handed from one gripper to the other.

- **SOP rule broken:** Steps 1 to 4. The left gripper works the left half. The right gripper works the right half. Nothing changes hands in the air.
- **Coaching note:** your half, your tools, your corners. Everything crosses on a rest, never in the air.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a wet band, a run, a bead, or a line on the glass, a corner still wet or streaked, water left on the frame, a tool off its place, the counter wet, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 5. Confirm the glass, the frame and the panel, the base, and the counter. Then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read the glass:** glare, or the reflection of the arms and the room, hides part of the panel, so whether it is clear cannot be judged.
- **Camera cannot read the glass-to-frame line**, so whether the blade ran onto the frame cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Panel not fixed:** a mirror that rocks, swings, or shifts on the wall under a correct pass.
- **Counter not fixed:** a counter that moves or rocks when a tool is set on it.
- **Defective squeegee:** rubber that is nicked, wavy, torn, or hardened, so a correct pass leaves a line whatever is done. Or a handle that will not hold in one gripper's opening.
- **Defective bottle:** one that will not spray a fan, that sprays a drifting mist however it is held, that leaks, that tips when gripped, whose body and trigger will not sit inside one gripper's opening, or whose trigger will not spring back on its own.
- **Defective cloth:** one that is torn, shedding onto the glass, already wet at episode start, that will not take the water off the blade in two strokes, or that will not keep its fold when held up by one edge, so it flops open or pulls out of the grip under a correct wipe.
- **Defective glass:** a chip, crack, scratch, or mark in the glass itself that a correct pass and a correct corner detail will not lift.

- **Cleaner fault:** a cleaner that dries to a film or leaves a haze on the glass however correctly it is sprayed and pulled.
- **Panel too large for one park:** a panel a correctly parked base cannot cover, so a corner, an edge, or the panel midline cannot be reached face-on without extending an arm.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so the bottle spot, the cloth rest, the squeegee rest, or part of that arm's half of the glass cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Take the spray bottle from the bottle spot
2. Fire one burst onto the left half of the glass
3. Fire one burst onto the right half of the glass
4. Return the spray bottle to the bottle spot
5. Take the squeegee from the squeegee rest
6. Take the cloth from the cloth rest and hold it in the wipe position
7. Set the blade at the top of a pass
8. Pull one top-to-bottom squeegee pass
9. Wipe the blade on the held cloth
10. Return the squeegee to the squeegee rest
11. Return the cloth to the cloth rest
12. Take the cloth from the cloth rest
13. Detail the top-left corner
14. Detail the bottom-left corner
15. Return the cloth to the cloth rest
16. Take the cloth from the cloth rest
17. Detail the top-right corner
18. Detail the bottom-right corner
19. Return the cloth to the cloth rest
20. Return both arms home and end the episode